

# Conditional Source-to-Directional-Support Composition

## Quantitative Interface Gates from Snapshot Balls to Support Cocycles

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### Abstract

We assemble the enclosure and viability layers into one conditional theorem. At level  $n$ , assume: an exact normalized source snapshot lies in an operator ball of radius  $\varepsilon_n$  around a computed snapshot; the computed packet has approximate spectral gap  $g_n > 2\varepsilon_n$ ; the induced packet and source radii propagate below the strict threshold-branch margin; the resulting exact branch admits outward-validated tail and normalized-base transitions

$$x_n \leq y_n, \quad y_{n+1} \leq F_{n,c_n}(y_n), \quad a_{n+1} \geq r_{n,c_n} a_n;$$

and, after a validated reset, the correlated multipliers

$$\kappa_n = r_{n,c_n} \frac{(1 - \sqrt{F_{n,c_n}(y_n)})_+^4}{(1 - \sqrt{y_n})^4}$$

have partial logarithmic sums bounded below by  $-C$ . Then the actual directional candidate satisfies

$$a_n(1 - \sqrt{x_n})_+^4 \geq a_N(1 - \sqrt{y_N})^4 e^{-C} > 0$$

for every later level. Four omission witnesses show that the gap, branch, outward-dominance, and cocycle packets cannot simply be deleted from this proof architecture.

An automated readiness audit verifies all eight RH-140–RH-147 archives and the RH-138 outward reference. The finite checkpoints are strong but not yet one end-to-end interval proof: 10/10 factorized source packets, 360/360 strict nominal branch margins, 330/330 outward residual pairs, 28/30 positive support tubes, and an 18/18 delayed clean suffix. Three all-level interfaces remain: uniform source/packet enclosure, propagation of packet balls through the thresholded recursion into the outward assembly, and a delayed support cocycle with uniformly bounded drawdown. Hence the theorem is a rigorous composition rule, not a claim that directional Stage A is closed.

## 1 The interface problem

The preceding papers establish exact local theorems, but a proof is only as strong as the interfaces between them. A spectral packet certified from one source approximation cannot be inserted silently into a later recurrence computed from unvalidated floating-point matrices.

Likewise, a finite positive directional floor cannot be promoted to an all-level liminf without a reset and a recurrent support estimate.

We therefore state the entire source-to-support route in one theorem. Its purpose is twofold: to make every quantitative handoff explicit and to separate a complete logical implication from the still-missing hypotheses. The perturbation ingredients are standard operator and interval methods [Kato, 1995, Bhatia, 2007, Moore, 1966]; the final controlled step uses the viability language developed in RH-144–RH-147 [Aubin, 1991].

Let  $S_n$  be the exact normalized source snapshot and  $\hat{S}_n$  its computed center. Let  $P_n$  and  $\hat{P}_n$  be the exact and computed rank-four packet projectors. The recursive packet update contains a clipped relative singular-value threshold, so its selected width is locally constant only inside a strict branch radius.

## 2 Quantitative gates

The first gate is the approximate spectral gap. Suppose

$$\|S_n - \hat{S}_n\| \leq \varepsilon_n \quad (1)$$

and the computed cluster-to-complement gap is  $g_n > 2\varepsilon_n$ . The RH-141 projector estimate gives

$$\|P_n - \hat{P}_n\|_{leqp_n} := \frac{\varepsilon_n}{g_n - \varepsilon_n}. \quad (2)$$

The strict factor two ensures the target cluster remains isolated on both sides of the perturbation.

Let  $m_n > 0$  be the exact threshold-branch radius from RH-143 and let

$$q_n = L_n^{(S)}\varepsilon_n + L_n^{(P)}p_n + \rho_n^{\text{round}} \quad (3)$$

be an outward propagated cross-matrix radius. The branch gate is  $q_n < m_n$ . Under this strict inequality, the exact and computed updates select the same width and clipping branch. Equality is deliberately excluded because it permits contact with the threshold surface.

On that branch, assume an outward residual calculation produces a monotone tail map  $F_{n,c}$  and a positive base-ratio lower  $r_{n,c}$  satisfying, for the actual matrices,

$$x_n \leq y_n, \quad y_{n+1} \leq F_{n,c_n}(y_n), \quad a_{n+1} \geq r_{n,c_n}a_n. \quad (4)$$

This hypothesis includes the source-to-assembly radii and both RH-138 Loewner residual guards; a nominal recurrence is not enough.

## 3 Conditional composition theorem

Write  $\phi(y) = (1 - \sqrt{y})_+^4$  and define the actual directional candidate

$$B_n = a_n\phi(x_n).$$

**Theorem 1** (Source-to-directional-support composition). *Suppose that after some level  $N$ :*

1. the source and packet gates (1)–(2) hold;
2. the strict branch gate  $q_n < m_n$  holds at every update;
3. the outward transition (4) holds for selected controls  $c_n$  and  $F_{n,c_n}(y_n) < 1$ ;
4. a validated reset gives  $a_N\phi(y_N) > 0$ ; and
5. for some finite  $C$  and every  $n > N$ ,

$$\sum_{j=N}^{n-1} \log \left( r_{j,c_j} \frac{\phi(F_{j,c_j}(y_j))}{\phi(y_j)} \right) \geq -C. \quad (5)$$

Then for every  $n \geq N$ ,

$$B_n \geq a_N\phi(y_N)e^{-C} > 0. \quad (6)$$

In particular the actual directional candidates have a positive eventual liminf.

*Proof.* The source gap and branch gates ensure that the outward transition applies to the actual recursively selected packet, rather than only to a nominal branch. Because  $x_n \leq y_n$  and  $\phi$  is decreasing,  $B_n \geq a_n\phi(y_n)$ . RH-147's sharp correlated multiplier and (4) give

$$a_{n+1}\phi(y_{n+1}) \geq r_{n,c_n} \frac{\phi(F_{n,c_n}(y_n))}{\phi(y_n)} a_n\phi(y_n).$$

Iterating and applying (5) proves (6). □

The theorem accommodates RH-145 delayed start: no hypothesis is needed on a finite prefix before  $N$ . It also allows local support losses, provided the signed cocycle drawdown remains bounded.

## 4 Four omission witnesses

Each interface carries genuine information.

**Proposition 2** (Architecture-relative necessity). *None of the gap, strict branch, outward-dominance, or cocycle packets can be removed from Theorem 1 while retaining its conclusion from the remaining scalar data alone.*

*Proof.* For the gap packet, a two-dimensional matrix with a double eigenvalue has no distinguished rank-one packet; arbitrarily small perturbations can rotate the chosen eigenspace. At  $g = 2\varepsilon$  the quantitative denominator reaches the separation wall.

For the branch packet, place a singular value exactly at the clipped relative threshold. Perturbations of either sign select different widths, so equality  $q = m$  is insufficient.

For outward dominance, let the nominal tail be zero but the actual tail equal one. The nominal support is positive while the actual support vanishes; no nominal calculation repairs the missing enclosure.

For the cocycle packet, take every multiplier equal to  $1/2$ . All local quantities are positive, yet the equality support trajectory is  $2^{-n}$  and has zero liminf. □

| RH  | layer           | finite checkpoint                          | missing all-level packet                    |
|-----|-----------------|--------------------------------------------|---------------------------------------------|
| 140 | source ball     | 10/10 rank-four radii below $10^{-3}$      | uniform normalized-source enclosure         |
| 141 | packet gap      | 4/10 universal; 10/10 quadratic diagnostic | interval-valid quadratic packet radius      |
| 142 | Arb packet      | 10/10 frozen binary packets                | uniform packet theorem                      |
| 143 | branch gate     | 360/360 strict nominal margins             | propagate packet balls through recursion    |
| 144 | control kernel  | 28/30 viable chains                        | all-level repeating/reset block             |
| 145 | delayed start   | 18/18 clean suffix                         | exclude recurrent future bad births         |
| 146 | base recurrence | 330/330 bounds, direct product lossy       | positive all-level base/cancellation packet |
| 147 | support tube    | 28/30 common positive tubes                | all-level lower-bounded support cocycle     |

Table 1: Readiness of the RH-140–RH-147 interface stack.

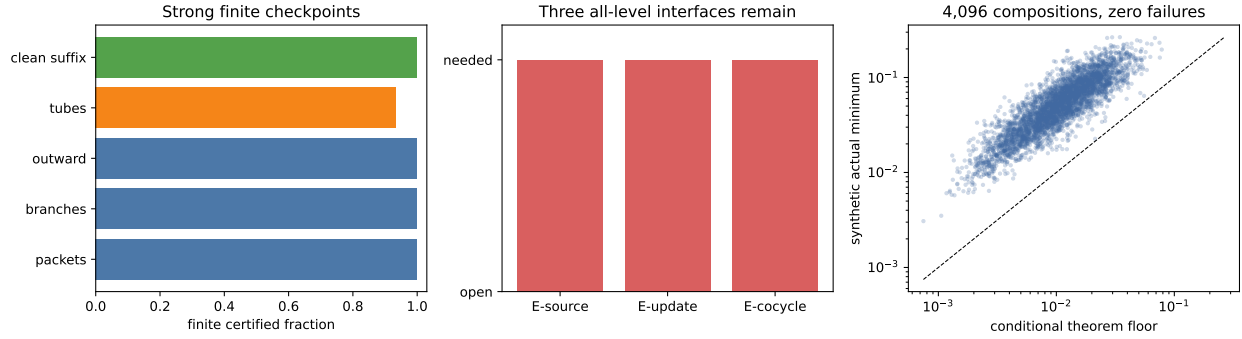


Figure 1: Strong finite checkpoints, the three open all-level interfaces, and the synthetic conditional-composition audit.

These witnesses are relative to this modular proof system. A different argument may replace a packet with stronger structural information, but it cannot simply omit the issue the packet controls.

## 5 Finite readiness audit

Table 1 records the strongest finite checkpoint and the corresponding open all-level interface. All eight archive-verification files and summary hashes pass; none asserts Stage A or the Riemann Hypothesis.

The RH-138 reference contributes 330/330 successful raw and bridge residual pairs, but its own claim boundary says that the reference assembly is not an interval source model. Thus the finite numbers cannot be multiplied into an end-to-end theorem. In particular, RH-142 packet balls have not yet been propagated through all RH-143 threshold updates into the exact Gram and tail matrices used by RH-138.

An independent 4,096-instance synthetic audit exercises the quantitative gap, branch, outward, and cocycle gates. Every actual synthetic support trajectory dominates the theorem floor. All four omission witnesses are also machine checked.

## 6 The reduced frontier and claim boundary

The proof architecture now has three open interfaces rather than a diffuse list of numerical observations:

$E_{\text{source}}$  : all-level source and packet balls

,

$E_{\text{update}}$  : branch-stable propagation into the outward assembly

,

$E_{\text{cocycle}}$  : delayed support cocycle with bounded drawdown

.

Once all three are proved on one common construction, Theorem 1 supplies eventual positive directional support without another conceptual step. The interfaces may be attacked separately, but their enclosures must be compatible.

RH-148 proves the conditional implication and audits finite readiness. It does not close a finite end-to-end interval source-to-support path, prove any of the three all-level interfaces, establish uniform Stage A, construct a Hilbert–Polya operator, identify zeta zeros, or prove the Riemann Hypothesis.

## References

Jean-Pierre Aubin. *Viability Theory*. Birkhauser, 1991.

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